Kei Hiroshima

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Citizenship: Japan

Research interests

I am interested in machine learning, optimization, and the intersection of the two. My PhD research focuses on two themes.
The first is continual learning. Machine learning models deployed in real-world settings must keep adapting to new tasks and data. However, training on a new task often causes catastrophic forgetting—the loss of previously acquired capabilities—while retraining from scratch to avoid it is costly. I study how to balance acquiring new knowledge with retaining existing knowledge, using model merging as my main approach.
The second is automating black-box optimization (BBO) with large language models (LLMs). BBO arises widely in practice, from hyperparameter optimization to experimental design, but applying it to a real problem requires designing a search space and selecting an optimizer, both of which demand expert knowledge. I formulate Black-Box Optimization Word Problems (BBOWP), a task in which an LLM makes these design choices from a natural-language problem description alone, and build a benchmark to evaluate LLMs on it.
I expect to complete my PhD in March 2028 and am seeking research positions in machine learning and optimization.

Machine learning, optimization, large language models, Continual learning, catastrophic forgetting, model merging, Black-box optimization, search space design, optimizer selection

Education

2025 – Present **Yokohama National University** – Yokohama, Japan
Ph.D. in Information Science, Graduate School of Environment and Information Sciences, Shirakawa/Uchida Laboratory (Supervisors: Prof. Shinichi Shirakawa, Asst. Prof. Kento Uchida).
Expected graduation: Spring 2028.

2023 **University of Technology Sydney (UTS)** – Sydney, Australia
Exchange Student, Faculty of Engineering and Information Technology
January – December 2023.

2022 – 2025 **Yokohama National University** – Yokohama, Japan
M.S. in Informatics, Graduate School of Environment and Information Sciences

2018 – 2022 **Yokohama National University** – Yokohama, Japan
B.E. in Electrical and Computer Engineering, College of Engineering Science Department of Mathematics, Physics, Electrical Engineering and Computer Science

Fellowships

2025 – 2028 **JST SPRING**
Support for Pioneering Research Initiated by the Next Generation (SPRING), Japan Science and Technology Agency (JST).

Research experience

2022 – Present **Graduate Researcher, Shirakawa/Uchida Laboratory** (Yokohama National University)
Supervisors: Prof. Shinichi Shirakawa, Asst. Prof. Kento Uchida.

Industry experience

Jan 2024 – Aug 2025 **AI Engineer Intern (Long-term)** — IDAJ Co., Ltd., Yokohama, Japan
Conducted machine-learning research to inform and prototype engineering solutions for industrial clients. The areas I researched were:

- Recognition models for 3D point-cloud data.
- Reinforcement-learning agents for energy distribution control.

Teaching experience

2025 – 2026 **Part-time Lecturer — Introduction to Mathematics, Data Science and AI**
Yokohama National University, Yokohama, Japan.

Publications

2026 **BBOWP-Bench: Evaluating LLMs on Black-Box Optimization Word Problems**
Y. Yamada, K. Hiroshima, N. Yoshinari, K. Uchida, and S. Shirakawa.
Advances in Neural Information Processing Systems (NeurIPS 2026), Sydney, Australia.
(Accepted, to appear)

2026 **Evaluating Large Language Models on Black-Box Optimization Word Problems**
Y. Yamada, K. Hiroshima, N. Yoshinari, K. Uchida, and S. Shirakawa.
34th FAN 2026, Kyoto, Japan.
[FAN Best Paper Award]

2026 **Tunable MAGMAX: Preference-Aware Model Merging for Continual Learning**
K. Hiroshima, K. Uchida, and S. Shirakawa.
Proceedings of the 28th International Conference on Pattern Recognition (ICPR 2026), Lyon, France.

2026 **Autoformulation Benchmark Suite of Black-Box Optimization Problems Described in Natural Language**
Y. Yamada, K. Hiroshima, N. Yoshinari, K. Uchida, and S. Shirakawa.
The 40th Annual Conference of the Japanese Society for Artificial Intelligence, 2026, Gunma, Japan.

2025 **Enhancing Atherosclerosis Risk Prediction with Strategic Feature and Case Selections in Large Language Model**
H. Murase, K. Hiroshima, K. Uchida, M. Ohashi, N. Kashihara, A. J. Viera, H. Daida, S. Shirakawa, and Y. Yano.
ESC Congress 2025, Madrid, Spain.

2025 **Controlling Forgetting via Model Merging in Continual Learning**
K. Hiroshima, K. Uchida, and S. Shirakawa.
39th SSI 2025, Iwate, Japan.
[SSI Excellent Presentation Award]

Honors and awards

2026 **FAN Best Paper Award**
FAN 2026 – The 34th Intelligent System Symposium, Fukuchiyama, Kyoto, Japan.
Awarded for the paper "Evaluating Large Language Models on Black-Box Optimization Word Problems."

2025 **SSI Excellent Presentation Award**
SSI 2025 – SICE Symposium on Systems and Information, Iwate, Japan.
Awarded for the presentation "Controlling Forgetting via Model Merging in Continual Learning."

Technical skills

Programming languages
Proficient in: Python, R

Frameworks and libraries
PyTorch, Keras, scikit-learn, OpenCV, Hugging Face

Tools and infrastructure
Git, Docker

Languages
Japanese (native), English (TOEIC 860 / IELTS 6.5: L 6.0 / R 7.0 / W 6.0 / S 6.0)